

THE LOFT

build manual

A five-sided timber platform 44 inches up, with a round hobbit door into the den underneath.

14 steps. Two saw angles. One weekend with a helper.

DECK

60 x 60 inch corner, five sides, 21.9 sq ft

PLATFORM HEIGHT

44 inches

GUARD

36 inches, balusters at 3 inches clear

ROUND DOOR

27 inch circle in a 30 inch square opening

SAW SETTINGS

45 degrees and 22-1/2 degrees. That is all.

POSTS

4 tall, 1 corner stub

SERVES

18 months to about 9 years



Generated from the parametric model. Every dimension traces to params.py.
Not engineer-stamped. Verify stud locations on site before drilling.

Before you start

TOOLS

Miter saw (10 inch is plenty)
Drill / driver, 1/4 and 9/16 sockets
Jigsaw
Router + circle jig (for the round door)
Stud finder
4 ft level
Speed square
Clamps
Orbital sander

THE FOUR RULES

- 1. The two wall ledgers carry the loft.**
The posts are secondary. Bolt the ledgers into studs, never drywall.
- 2. Everything is screwed, not nailed.**
- 3. No guard gap over 3-1/2 inches,**
anywhere, ever. Check with a block.
- 4. Cut one, test fit, then cut the rest.**

KEEPS THE LITTLE ONE SAFE

The ladder LIFTS OFF. Two pins. That is what keeps an 18-month-old off a 44 inch platform, not the gate.

No rope anywhere a child can loop it.
No shelves in the guardrail - they are a ladder.
No lock on the round door, ever.
A 2 inch mat under the open faces.

Cut everything first

pick them off as you go - it is much faster than cutting as you build

☐ 2x 1x3
☐ 2x 2x2
☐ 18x 2x2
☐ 4x 2x2
☐ 2x 2x4
☐ 1x 2x4
☐ 2x 2x4
☐ 1x 2x4

1'-11-1/2"
3'-5-7/16"
2'-7"
3-3/4"
4'-0-9/16"
3'-6-7/16"
3'-1-15/16"
3'-1"

☐ 2x 2x4
☐ 9x 2x4
☐ 4x 2x4
☐ 4x 2x4
☐ 5x 2x4
☐ 3x 2x4
☐ 1x 2x4 PT
☐ 2x 2x4 PT

3'-0"
2'-10-3/4"
2'-6"
2'-1-1/2"
1'-3"
4-3/4"
3'-6-7/16"
2'-6"

☐ 1x 2x6
☐ 1x 2x6
☐ 1x 2x6
☐ 1x 2x6
☐ 2x 2x6
☐ 2x 2x6
☐ 1x 2x6
☐ 1x 2x6 PT

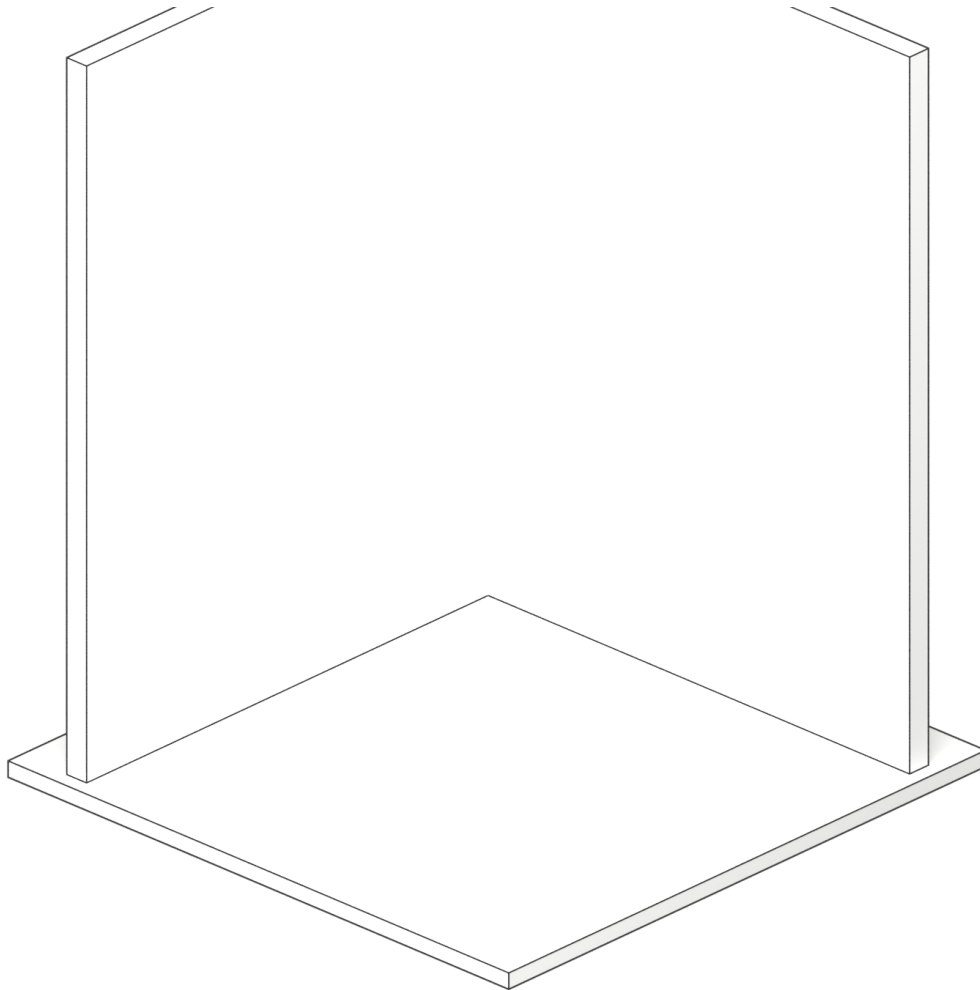
4'-9"
4'-7"
3'-6-7/16"
3'-3"
2'-6"
2'-1-7/16"
2'-0"
5'-0"

☐ 1x 2x6 PT
☐ 2x 3/4" ply
☐ 4x 3/4" ply
☐ 4x 4x4 PT
☐ 1x 4x4 PT

4'-10-1/2"
1'-11-1/2"
8"
6'-8"
3'-1-3/4"

1

Find your studs



YOU NEED

stud finder, pencil, 4 ft level

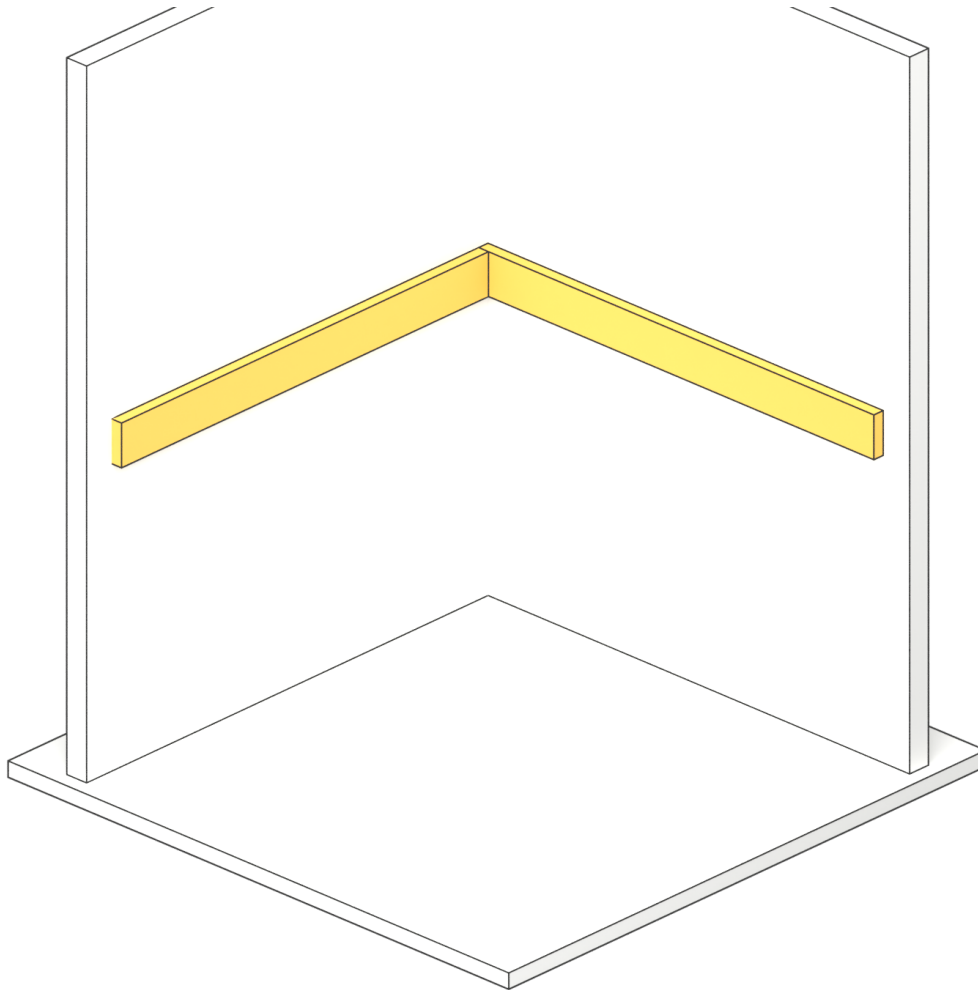
Mark the centre of every stud on both walls, floor to 48 inches.
Mark a level line at 37-3/4 inches on both walls. That is the underside of the deck framing and every other height comes off it.
If your studs are not 16 inches on centre, that is fine. Move the bolts to land on wood. Never anchor a ledger to drywall.

! Do not skip this. Everything above hangs off these two walls.

amber = install this step grey = already built

2

Bolt the two wall ledgers



YOU NEED

drill, 9/16 socket, 3/8 x 4 lag screws + washers

PARTS FOR THIS STEP

1x	2x6 PT	5'-0"
1x	2x6 PT	4'-10-1/2"
8x	3/8" x 4" lag screw + washer	
1x	tube construction adhesive (optional)	

North wall ledger first, west wall ledger butts into it.

Top edge of both ledgers sits on your 37-3/4 inch line.

Two 3/8 x 4 lag screws into EVERY stud, staggered high and low.

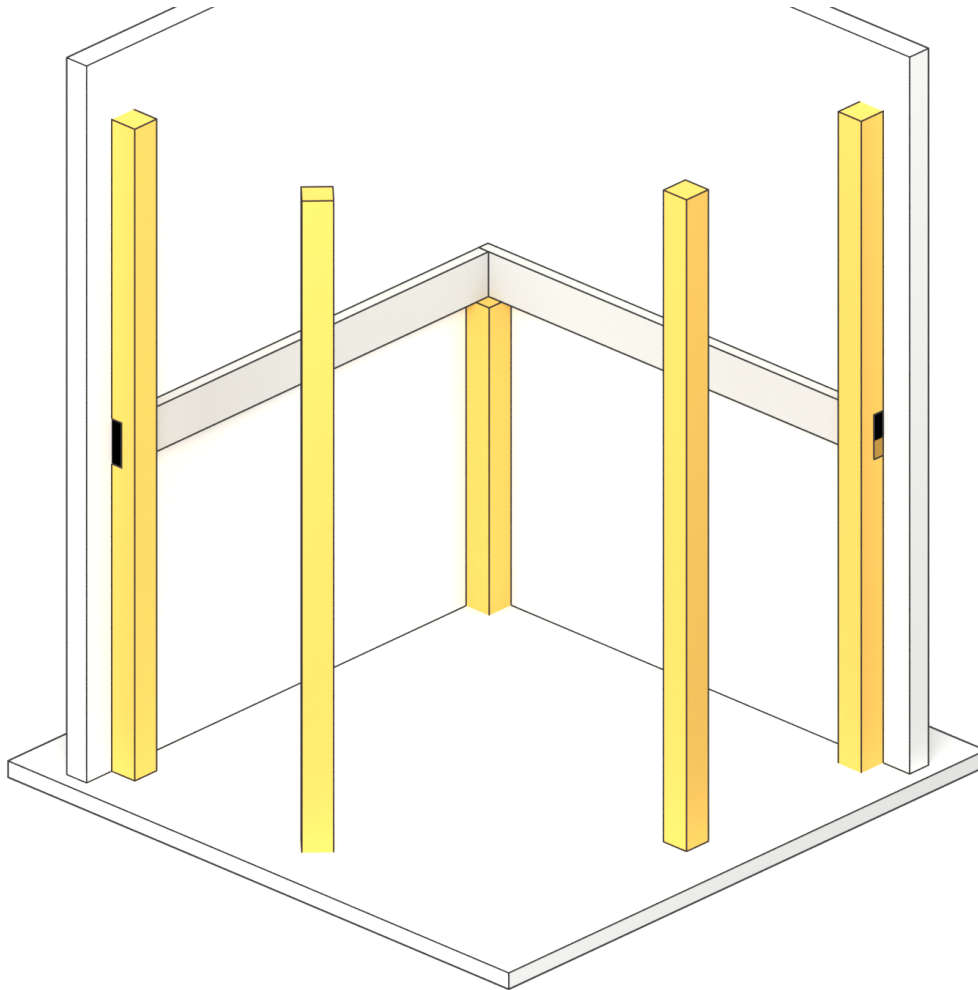
Pre-drill a 1/4 inch pilot so the lag does not split the ledger.

! These two boards carry the whole loft. The posts are secondary.

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3

Stand the five posts



YOU NEED

post bases, level, 1/4 x 3 lag screws

PARTS FOR THIS STEP

1x	4x4 PT	3'-1-3/4"
4x	4x4 PT	6'-8"
3x	Simpson ABU44Z post base	
12x	1/4" x 3" lag screw	

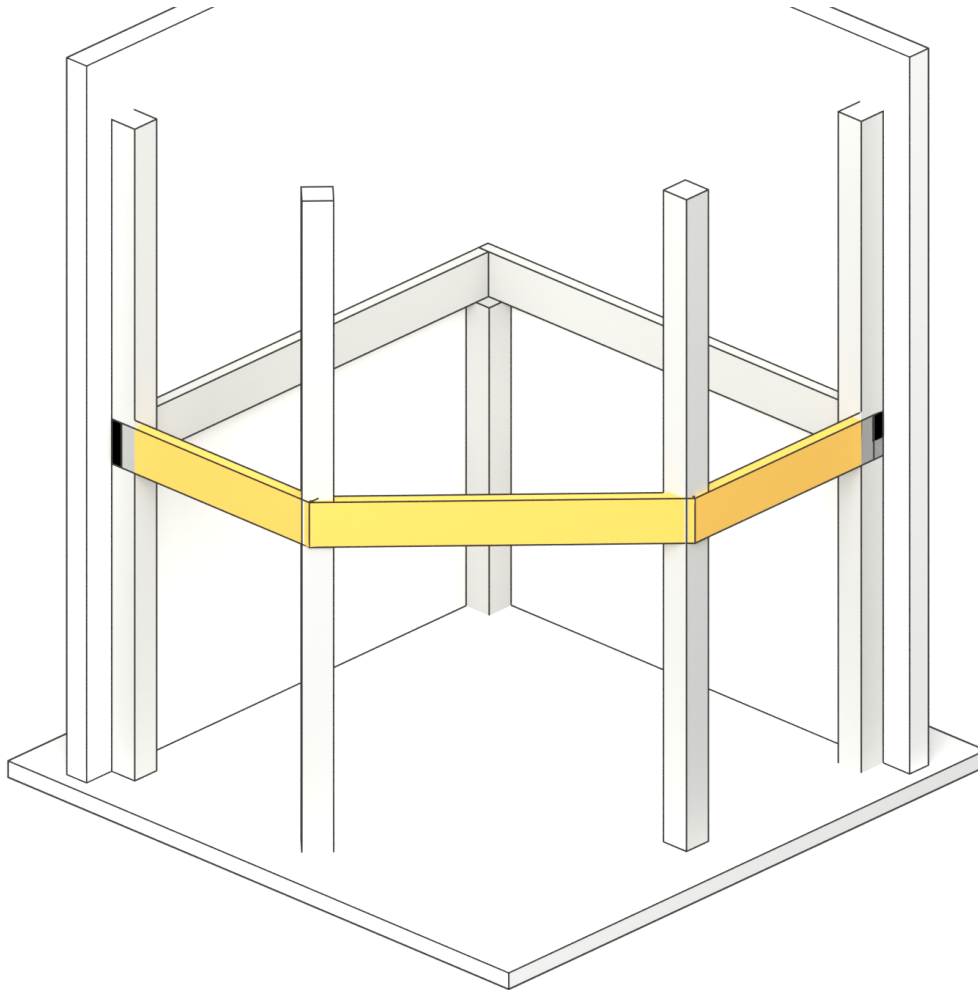
Four tall posts at 80 inches: B, C, D, E.
 One short stub at 37-3/4 inches in the room corner: A.
 Post D is the diagonal corner. Turn it 45 degrees so its faces meet both rim boards square.
 Plumb every post in two directions before you fix the base.

! Post B and post E sit against a wall. Lag them to a stud as well.

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4

Rim boards — only two saw settings



YOU NEED

miter saw, 3 inch structural screws

PARTS FOR THIS STEP

2x	2x6	2'-6"
1x	2x6	3'-6-7/16"

Three boards close the deck: 30 inches, 42-7/16 inches, 30 inches.

Set the saw to 45 degrees for the two ends that meet a wall.

Set it to 22-1/2 degrees for the four ends at the diagonal corners.

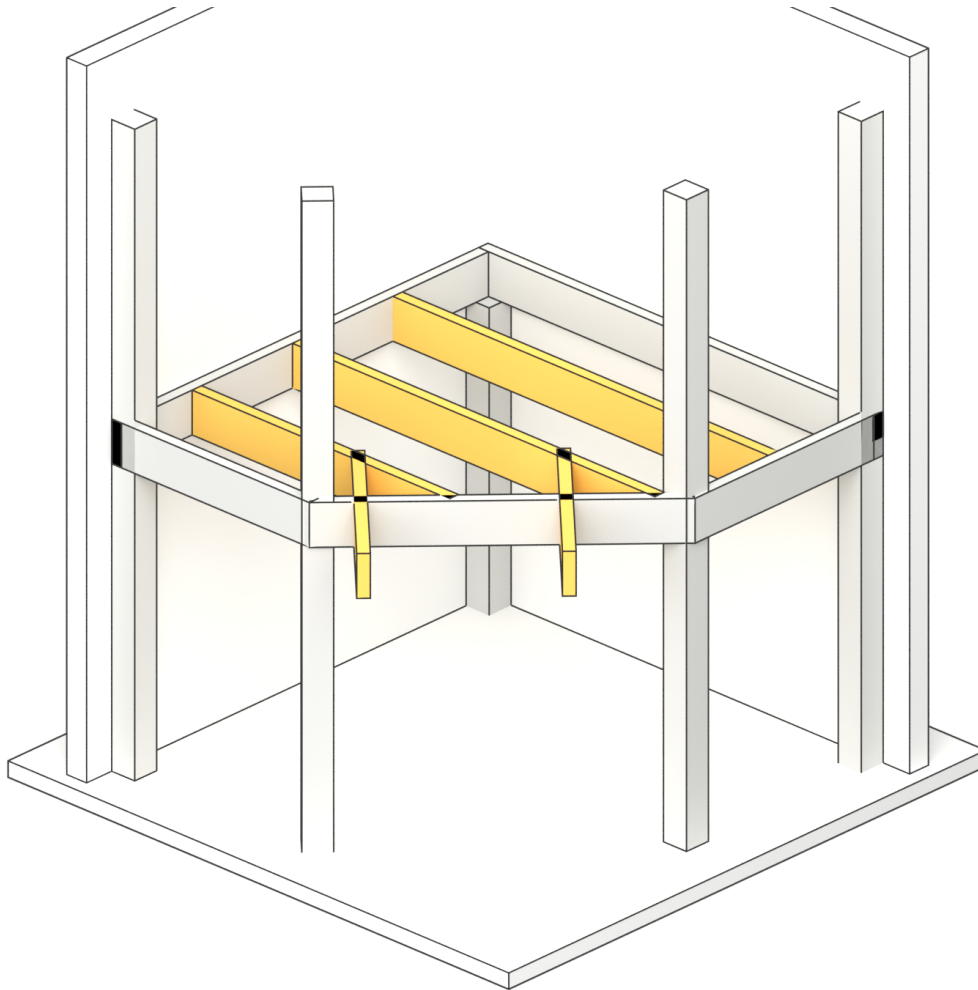
That is every angle cut in this entire build.

! Cut the 42-7/16 board last and measure the real gap first.

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5

Hang the joists



YOU NEED

joist hangers, hanger nails, speed square

PARTS FOR THIS STEP

1x	2x6	4'-9"
1x	2x6	4'-7"
1x	2x6	3'-3"
2x	2x6	2'-1-7/16"
3x	Simpson LUS26 joist hanger	
4x	Simpson LSU26 skewable hanger	
1x	box hanger nails	

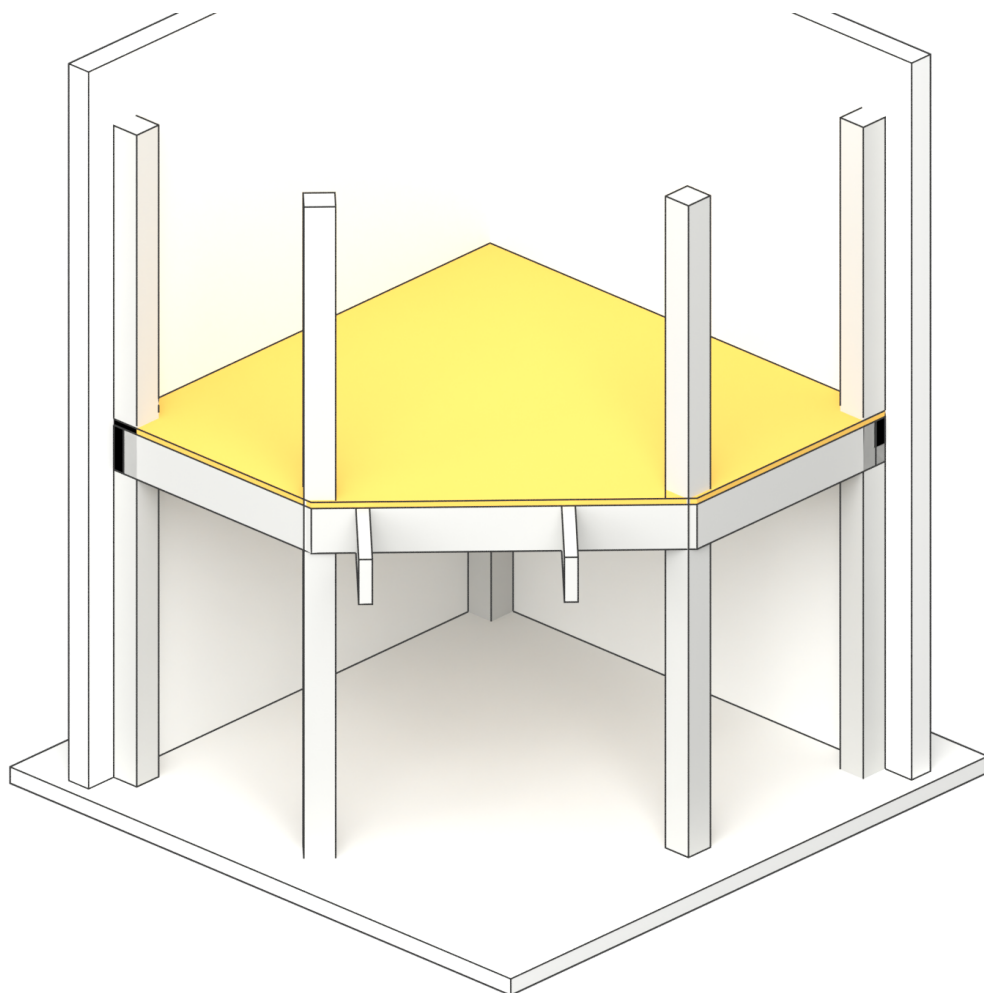
Three 2x6 joists at 16 inches on centre, hung off the west ledger. J1 is square both ends and takes plain LUS26 hangers. J2 and J3 die into the diagonal rim at 45 degrees. Use the adjustable LSU26 hangers and bend them to the angle. Two short blocks behind the diagonal support the deck edge.

! Every hanger hole gets a nail. Do not toe-nail a joist.

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6

Screw down the deck



YOU NEED

jigsaw, chalk line, 1-5/8 inch screws

PARTS FOR THIS STEP

1x 3/4" plywood deck, five-sided, ~22 sq ft

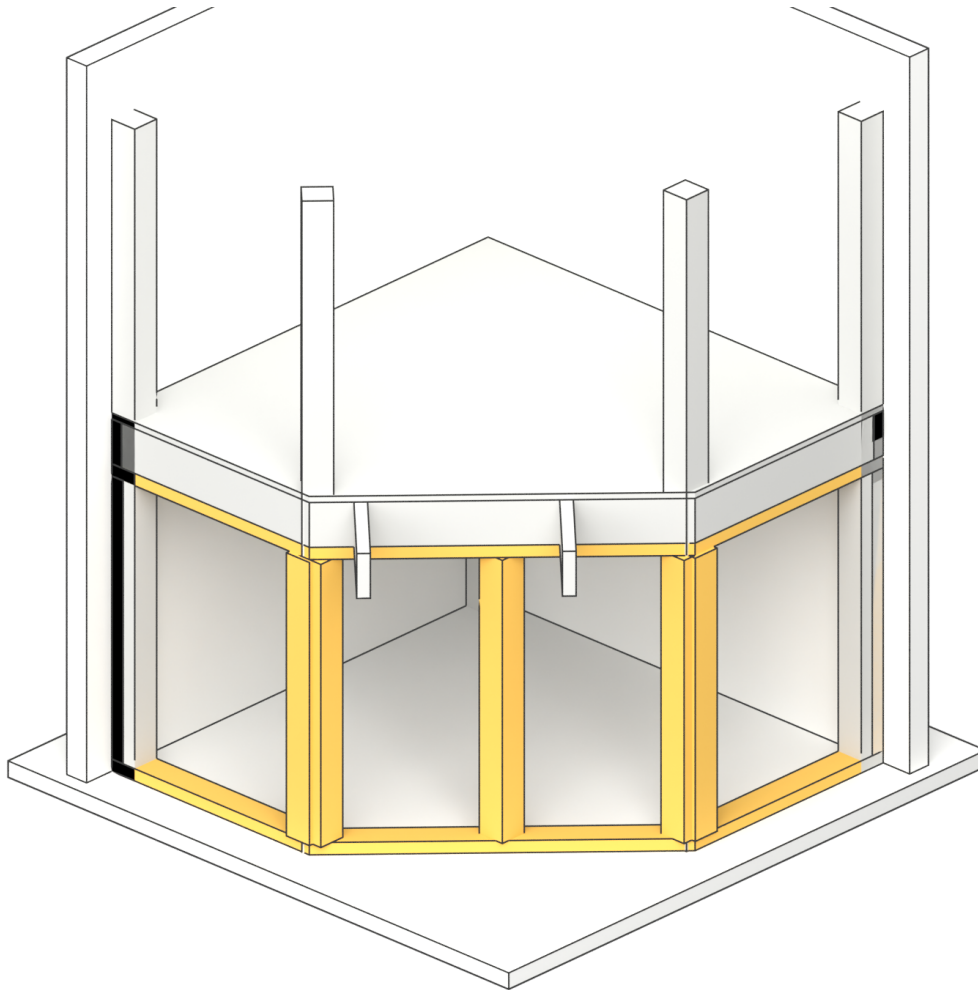
1x box #8 x 1-5/8" screws

One piece of 3/4 inch plywood, cut to the five-sided outline. Trace it off the frame you just built rather than off the drawing. Screws every 8 inches into every joist and all the way round the rim. Countersink every head. Bare feet and bare knees live up here.

! Screwed, never nailed. Nails work loose and back out.

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7 Frame the knee wall



YOU NEED

nail gun or screws, level

PARTS FOR THIS STEP

2x	2x4 PT	2'-6"
2x	2x4	2'-6"
7x	2x4	2'-10-3/4"
1x	2x4 PT	3'-6-7/16"
1x	2x4	3'-6-7/16"

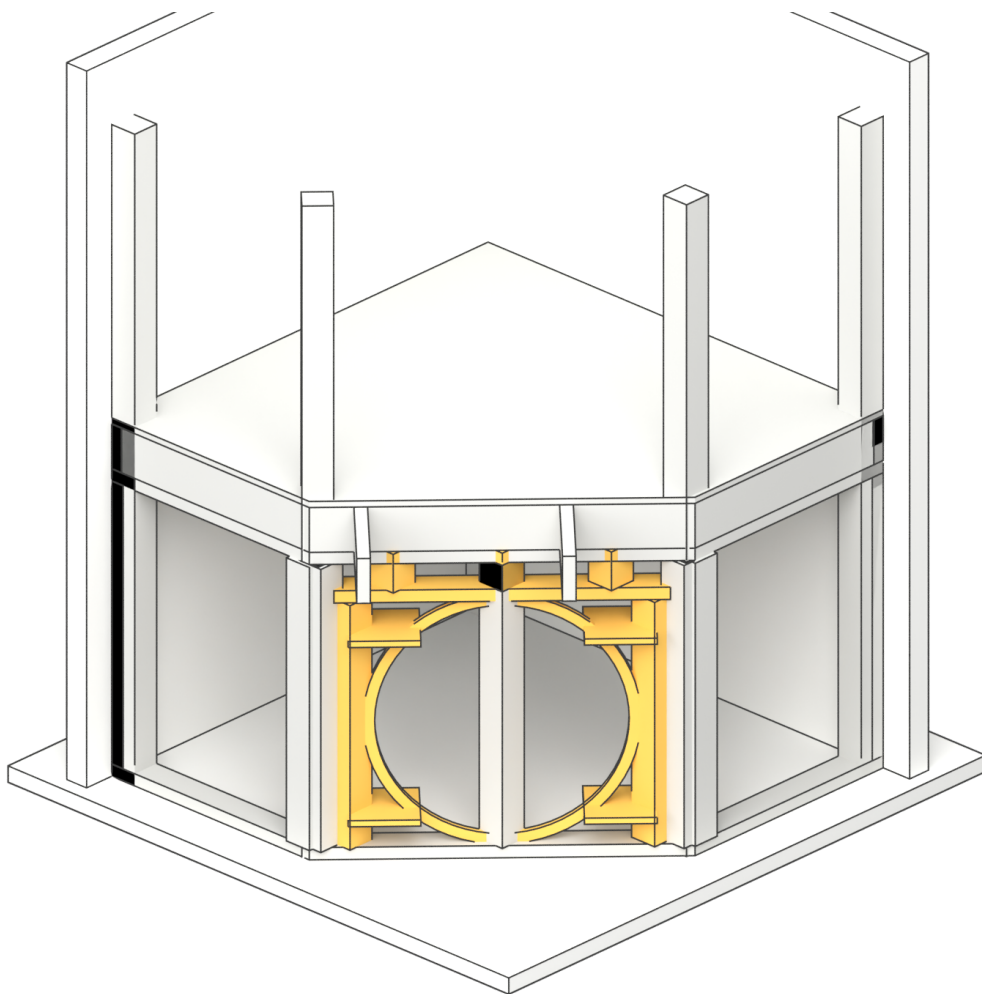
Bottom plate is pressure treated because it touches the subfloor. Studs are 34-3/4 inches. Cut one, hold it in place, confirm, then cut the rest.

This wall carries nothing. It encloses the den and holds the door.

! Leave the door bay open. The next step fills it.

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8 Frame the round door opening



YOU NEED

router + circle jig, jigsaw, 3/4 inch plywood

PARTS FOR THIS STEP

2x	2x4	2'-6"
2x	2x4	2'-10-3/4"
1x	2x4	3'-1"
3x	2x4	4-3/4"
4x	3/4" ply	8"
1x	3/4" ply ring, 30" outside / 28" inside	
1x	router circle jig	

Frame a SQUARE 30 x 30 inch rough opening. King studs, jack studs, a flat 2x4 header and three short cripples above it.

You never cut a curved stud.

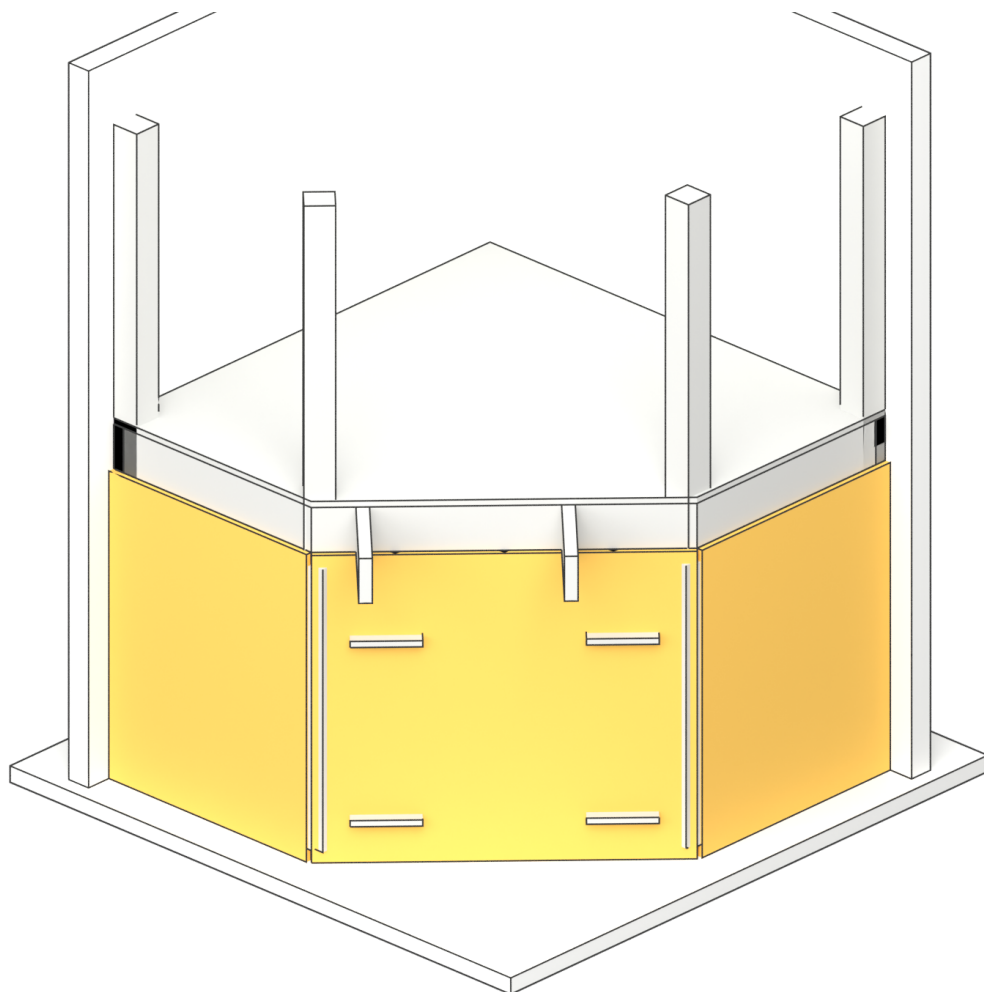
Then cut four plywood quadrants with a router and circle jig and screw them into the corners of that square. They form the 28 inch circle. Sand the joints until you cannot feel them.

! Dry-fit all four quadrants before you glue anything.

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9

Skin it and set the porthole



YOU NEED

jigsaw, hole saw, 1-5/8 inch screws

PARTS FOR THIS STEP

3x 3/4" ext ply skin panel

1x 8" acrylic porthole disc

1x clear silicone

Three plywood panels close the knee wall.

SCREW the panel on the north face, never glue it. That panel is how an adult gets in if a small child is hurt inside.

Cut the 8 inch porthole with a hole saw and bed the acrylic disc in clear silicone.

Drill a pattern of 1-1/4 inch holes near the top of two panels so the den breathes.

! 1-1/4 inch holes. Never 3/8 to 1 inch - that traps fingers.

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10 Build the guards



YOU NEED

miter saw, spacer block, 2-1/2 inch screws

PARTS FOR THIS STEP

4x	2x4	2'-1-1/2"
18x	2x2	2'-7"
2x	2x4	3'-1-15/16"
2x	2x4	3'-0"

Toe board flat on the deck first, then the top rail at 80 inches, then the balusters between them.

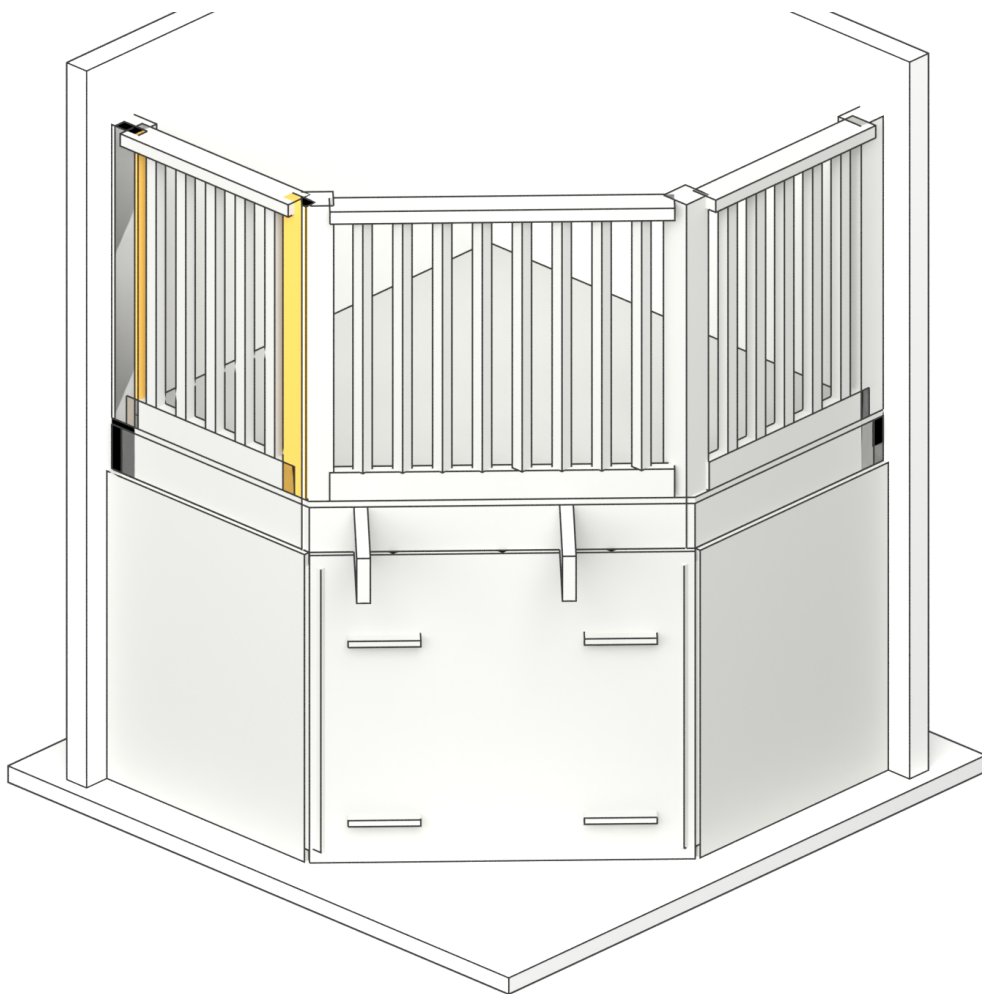
Cut ONE spacer block and use it for every gap. Do not measure each one; you will drift.

Gaps come out between 2-7/8 and 3 inches.

! Check every finished gap with a 3-1/2 inch block. Nothing passes.

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11 Hang the gate



YOU NEED

self-closing spring hinges, gate latch

PARTS FOR THIS STEP

- 1x pair 3 inch self-closing spring hinges
- 1x gravity gate latch, black
- 1x magnetic catch

The gate leaf is built like a guard panel, 19-1/2 inches wide. Hinge it so it swings IN over the deck, never outward. Self-closing spring hinges, a gravity latch and a magnetic catch. No lock, no bolt, nothing a child can shut themselves behind.

! Let it go from wide open ten times. It must latch every time.

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12 Build the ladder — it lifts off



YOU NEED

dado blade or router, 3/8 x 5 hitch pins

PARTS FOR THIS STEP

1x	2x6	2'-0"
2x	2x4	4'-0-9/16"
2x	2x2	3'-5-7/16"
4x	2x2	3-3/4"
5x	2x4	1'-3"
2x	3/8" x 5" hitch pin with lanyard	
1x	2x6 hanger rail (cut at step 4)	

Two 2x4 stringers, five rungs at 7-5/16 inch risers.

Dado each rung 3/4 inch into the stringers, then two screws per end.

Notch the top of each stringer to hook over the 2x6 hanger rail.

A 3/8 inch pin through each stringer locks it down.

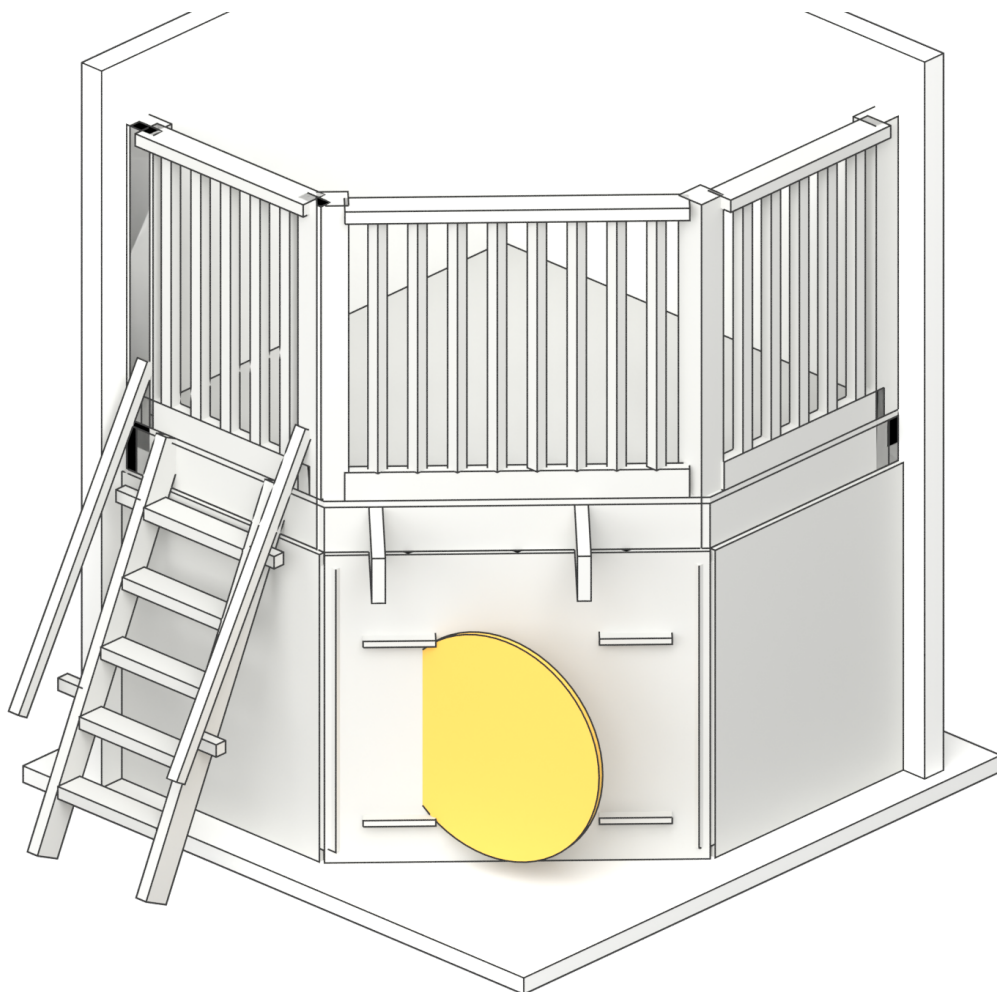
Rigid 2x2 grab rails on both sides. No rope.

! Pull two pins and the loft is unreachable. That is the toddler lock.

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13

Den shelf and the door itself



YOU NEED

strap hinges, 1/4 inch carriage bolts

PARTS FOR THIS STEP

2x	3/4" ply	1'-11-1/2"
2x	1x3	1'-11-1/2"
1x	27" circle, 3/4" ply - the door slab	
2x	12" black strap hinge	
1x	2-3/4" round wood knob	
1x	magnetic catch	
1x	door stop	

Two low shelves inside the den for board books.
The door slab is a 27 inch circle of 3/4 inch plywood, about 9 lb.
Do not build it thicker. A toddler swings this.
Through-bolt both strap hinges with carriage bolts. Screws sag.
Add a doorstop so it cannot swing hard onto fingers.

! Magnetic catch only. It must push open from the inside.

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14

Sand, finish, and put the mat down



YOU NEED

150 and 180 grit, zero-VOC primer, mat

PARTS FOR THIS STEP

- 1x 2 inch folding gym mat, 4 x 8 ft
- 1x qt zero-VOC primer
- 1x qt deep green eggshell
- 1x qt water-based satin topcoat

Sand every surface a child can grip to 180. Balusters, rails, rungs, grab rails, door edges, the whole den interior. Run a bare forearm over everything below 40 inches. Rough-sawn texture only where nobody touches it. Zero-VOC primer, then paint and topcoat. A 2 inch mat goes under the ladder and the open faces.

! 3/4 inch foam is not enough for a 44 inch fall.

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Done.

Now it has to survive being lived with.

18 months - 3 years

Den and the round door only. Ladder stored flat, off the structure. Nothing to fall from.

3 years +

Hang the ladder. Loft in use. Gate self-closes. Only once he climbs it unassisted, both ways.

5 years +

Unscrew the removable panel on the north face and add the rope bridge to a second structure.

CHECK THESE, ON A SCHEDULE

30 days: re-torque every lag. Green lumber shrinks and they WILL be loose.

6 months: lags again, gate latches, ladder pins present, no gap grown past 3-1/2 inches. Write the date on the ledger in pencil.